Caterpillar Performance Handbook

CATERPILLAR®

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CATERPILLAR PERFORMANCE HANDBOOK

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Performance information in this booklet is intended for estimating purposes only. Because of the many variables peculiar to individual jobs (including material characteristics, operator efficiency, underfoot conditions, altitude, etc.), neither Caterpillar nor its dealers warrant that the machines described will perform as estimated.

NOTE: Always refer to the appropriate Operation and Maintenance Manual for specific product information.

Materials and specifications are subject to change without notice.

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PUBLICATION UPDATE:

This publication removes main technical specifications from Performance Handbook version 49. For accurate reference of Specifications, Rimpull and Retarding Charts, Dimensions, Matching Guides, etc. please reference published machine specific documents such as Product Brochures and Technical Specifications through www.cat.com or your Caterpillar Dealer.

The Cat Product Line Section only includes production models at the moment of publishing this book.

PREFACE

Machine performance must ultimately be measured in unit cost of material moved, a measure that includes both production and costs. Factors bearing directly on productivity include such things as weight to horsepower ratio, capacity, type of transmission, speeds and operating costs. The Performance Handbook considers these factors in detail. There are other less direct machine performance factors for which no tables, charts or graphs are possible. Serviceability, parts availability and operator convenience are examples. In comparing machine performance, all factors should be considered. This Handbook is intended as an aid which, when coupled with experience and a good knowledge of local conditions, can assist in estimating true machine performance.

Many sections of the Handbook include tables or curves showing cycle times or hourly production figures for Cat® machines under certain conditions. Statements of conditions always accompany or precede the curves or tables. Before using any performance information in this Handbook, a complete understanding of the qualifying conditions is essential. The data is based on field testing, computer analysis, laboratory research and experience; and every effort has been made to assure their correctness.

However, all such data is based upon 100% efficiency in operation — a status which cannot be achieved continuously even under ideal conditions. Thus, in using such performance and production data, it is necessary to correct the results indicated in the handbook tables by appropriate factors. This allows for the anticipated actual job efficiency, operator efficiency, material characteristics, haul road conditions, altitude and other factors which may reduce performance or production on a particular job.

Methods for estimating machine owning and operating costs vary widely, depending on locality, industry practices, owner preferences and other factors. One method is suggested in the Handbook section on Owning and Operating Costs. When used with good judgment, it has provided reasonably accurate estimates in the past. Included in the Owning and Operating Section are guidelines, based on working conditions, to assist in estimating consumption of fuel and lubricants, tire life and repair costs for Cat machines. However, what one Handbook user regards as "excellent" conditions, another may consider "severe" or "average", depending on his own experience and basis of comparison. Therefore, these guidelines should be considered only approximations.

Caterpillar has made every effort to assure that the information contained in this Handbook is accurate and is a fair statement of the results to be achieved in the circumstances indicated. However, because of the many variables involved in estimating the production or performance of earthmoving machinery, their consumption of fuel and lubricants, tire life and repair costs, and the possibility of inadvertent errors or omissions in assembling this data, Caterpillar cannot and does not imply that all data in this book are complete nor that this level of performance will be achieved on a given job.

Caterpillar

ENVIRONMENT AND SUSTAINABILITY

Caterpillar supports environmental responsibility through sustainable development. Our products and services are intended to support sustainable development of global resources and they will meet or exceed applicable regulations and standards wherever they are initially sold. We establish and adhere to environmentally sound policies and practices in product design, engineering, and manufacturing. We educate and encourage our customers to use the products they purchase from us in environmentally responsible ways. We take effective steps to continually increase the natural resources efficiency and cleanliness of our facilities. When available, the various product groups have included relevant sustainability data with their content updates for inclusion in the Performance Handbook.

OPERATOR AND MACHINE PROTECTION

A well trained operator, working under suitable conditions, utilizing a modern, properly-equipped machine provides a machine-operator team capable of giving maximum production. These factors, along with appropriate job site rules and communication procedures, are essential to coordinate people and machines working together.

Appropriately protected and maintained machines are less likely to suffer premature component failure or damage, and give operators the confidence and assurance they need to carry out their work. Furthermore, training is not complete until the operator reads, understands and agrees to follow the instructions provided in the Operation and Maintenance Manual included with every Cat machine.

Employers have a duty to provide a safe work place for their employees. The purchaser of a Cat machine has a duty to review his/her particular application and job site for the machine to identify potential hazards inherent to that application or job site. Based on the results of this hazard analysis, the appropriate operator and machine protection configuration can be determined.

Caterpillar designs, builds, and tests its products to ensure the safety of operators, maintenance persons, service persons, and bystanders. That means people in, on and around Cat products. Caterpillar provides as standard equipment the appropriate operator and machine protection for most applications. However, particular applications, including the use of some Work Tools, may require additional operator and/or machine protection. Caterpillar offers related options for most such applications. However, there may be very special applications where the Cat dealer or the Purchaser may want to fabricate, or request Caterpillar to provide, custom or special guarding. Your Cat dealer can help you with this hazard analysis and guarding configuration process.

I. Operator Training and Protection Practices

Remember that any kind of machine or mechanical device can be hazardous if not kept in good condition, or if operated by careless or improperly trained operators, or if operated in an irresponsible manner.

Listed below are some recommended basic steps that can be broadly applied to most work environments:

- Train operators for the job they are assigned to do. The length and type of training must comply with governmental and local regulations wherever they apply. As an example, machine operators in mining activities must be trained in accordance with Mine Safety and Health Administration (MSHA) regulations. Where specific regulations do not apply, no operator should be assigned to a job until he or she meets the following minimum requirements:
  - Completes proper training to operate the assigned machine and understands that seat belts must be worn whenever seated in operator's compartment. SEAT BELTS SAVE LIVES!
  - Reads and understands the Operation & Maintenance manual for that machine, and knows that a copy of that manual is stored in the operator's compartment.
  - Reads and understands the AEM (Association of Equipment Manufacturers) Safety Manual, or any other furnished manual related to rules for safe machine operation and identification of hazards. For example, that includes the Work Tool Operation and Maintenance Manual if a Cat Work Tool is involved in the given application.
  - Has appropriate personal protective equipment and knows how to use it. This includes such things as hard hat, gloves, safety glasses, hearing protection, high-visibility vest, and safety shoes.
  - Knows what the job requirements are, what other machines are working in the area, and is aware of any hazardous conditions that may arise.

- Be sure operators are alert and in proper physical and mental condition to perform their work assignments safely. No machine should be operated by a person who is drowsy, under the affect of medicines or drugs, suffers blackouts, or is suffering from any physical or mental distraction that could contribute to unsafe operation.
- Maintain proper job conditions and working procedures. Check the job for possible hazards, both above and below ground level. Look for all possible sources of danger to the operator and others in the area. When operating in hazardous conditions the door and windows must always be closed. Pay particular attention to conditions which may be hazardous or near the operating limits of the machine: e.g., side slopes, steep grades, potential overloads, etc. Examine the work site for restricted traffic patterns, obstructed views, congestion, underground power or gas lines, etc. If the machine is equipped with a Quick Coupler, always make sure the Work Tool is properly attached by conducting an attachment test as directed in the Quick Coupler or Machine Operation and Maintenance Manual. Hazardous work conditions should be corrected wherever possible and adequate warnings should be posted when applicable.
- Provide the correct machine to handle the job and equip it properly for the job to provide the necessary operator protection. Check for compliance with all applicable governmental and local regulations. It is the legal responsibility of the machine owner or employer to see that his equipment complies with, and is operated in accordance with, all such requirements.
- Make sure the machine is properly maintained. The operator at the beginning of each shift should perform a walk-around inspection before the machine is placed in operation. This process is described in the machine and Work Tool Operation and Maintenance Manual. If this inspection reveals any problems that could affect safety, the machine or Work Tool must not be operated until these problems are corrected. Some examples include:
  - Loose, bent or missing grab irons, railings or steps;
  - Worn, cut or missing seat belts (any seat belt over three (3) years old must be replaced regardless of condition);
  - Damaged windows in the operator's compartment;
  - Worn, rubbing or abraded electrical insulation and hydraulic hoses;
  - Material or unwanted debris accumulation;
  - Incompatibility of the component attachments (Quick Couplers);
  - Hydraulic leaks that could impair the lock/secure feature of a Quick Coupler or other securing devices;
  - Any fluid leaks; and
  - Missing or damaged guards.
- Know the limits of your machine and equipment. With certain Work Tool combinations, including Quick Couplers, the Work Tool can hit the cab or the machine. Always check for interference limits when first operating.
- It is the machine owner's or employer's responsibility to ensure the machine is properly maintained. Your Cat dealer will be glad to assist you in selecting and equipping the machine best suited for your job and in providing maintenance for your machines.

II. Machine Modifications

Modifications must not be made to the machine that:

- Interfere with operator visibility;
- Interfere with ingress or egress from the machine;
- Exceed the rated payload or gross combination weight of the machine resulting in overloading the braking and/or steering system or the roll-over protective structure (ROPS) capacity rating (shown on a plate affixed to the ROPS); or
- Place objects in the cab that intrude into the operator's space or that are not firmly fixed into place.
- Are not authorized by Caterpillar.

III. Operator-related Equipment Options

Each job presents unique conditions that must be taken into account. Consider direct dangers to the operator as well as all possible sources of distraction that could reduce operator efficiency and increase the chances of costly and dangerous mistakes. Climate-controlled, sound-suppressed cabs, and special exterior lighting are options available from Caterpillar that can address requirements of special working environments.

"Flexible" machines include hydraulic excavators (track-type, wheel-type, and compact), skid-steer loaders, backhoe loaders and integrated tool-carriers can utilize interchangeable "Work Tools" to accomplish specific tasks. Work Tools or any tool used in hazardous applications like demolition, quarry, logging, stump grinding, scrap handling, milling, and scaling, can create a need for special operator guarding. When flying debris from impact, cutting, shearing or sweeping attachments is present, additional protective devices such as a front screen, Falling Object Guarding System (FOGS, includes top & front guarding), thick polycarbonate windshields or a combination of these is recommended by Caterpillar. The failure to provide proper operator/machine guarding in some of these applications can lead to machine damage, personal injury or death. Contact your Cat dealer for operator guarding options on your machine.

IV. Machine Protection

Check the job for unusually demanding conditions that could cause premature failure or excessive wear of machine components. Additional protective devices such as heavy-duty radiator guards, crankcase guards, engine enclosures, track roller guards and/or brake shields may be needed. Also, consider the use of anti-vandalism devices, such as cap locks and instrument panel guards. The failure to provide proper guarding in some of these applications may lead to machine damage, personal injury or death. Contact your Cat dealer for machine-protection and vandalism-prevention options for your machine.

V. Fire Prevention

Remember that most fluids on your machine are flammable!

To minimize the risk of fire, Caterpillar recommends following these basic steps:

- Always perform the Walk-Around Inspection described in Part I. It can identify many of the fire hazards described below.
- Remove trash (leaves, twigs, papers, etc.) that may accumulate in the engine compartment or around other hot parts on the product.
- Do not operate a machine if leakage of flammable fluids is noticed. Repair leaks before resuming machine operation. Most fluids used in Cat machines should be considered flammable.
- Keep access doors to major machine compartments in working order to permit the use of fire fighting equipment, should a fire occur.
- Avoid attaching electrical wiring to hoses and tubes that contain flammable or combustible fluids. Hydraulic hoses can move during machine operation and abrade wires and other hoses if improperly secured.
- Replace any rubbing, damaged, frayed, kinked or leaking hydraulic hoses or fittings. Use genuine Cat parts or their equivalent, including both pressure and temperature limit capabilities.
- Follow safe fueling practices as described in Cat Operation and Maintenance Manuals, AEM Safety Manuals, and local regulations. Never store flammable fluids in the machine operator's compartment, nor smoke while fueling the machine.
- As an additional safety measure, keep a fire extinguisher on the machine in a location as specified in the Operation and Maintenance Manual.
- Consider installation of an after-market fire-suppression system (FSS) on the equipment if the application and working conditions warrant it.

VI. Safety Regulations

Regulations vary from country to country and often within country. Your Cat dealer can assist you in properly equipping your machine to meet applicable requirements. **Note: The general summaries given below are not substitutes for Owners or Employers reading and being familiar with the appropriate local laws.**

(a) United States (US)

With a few exceptions, all machine operations in the United States are covered by federal and/or state regulations. If the machine is used in mining activities, the regulations are administered by the Mine Safety and Health Administration (MSHA). Other activities, including construction, are under regulations administered by the Occupational Safety and Health Administration (OSHA). These agencies require employers to provide a safe working environment for employees. Caterpillar has the same objective.

OSHA and MSHA have adopted criteria for ROPS, Falling Object Protective Structures (FOPS), seat belts, warning horns, back-up alarms, operator sound levels, steering systems, and braking systems. Additional operator's compartment protection may be required for machines engaged in logging, demolition and other special applications.

(b) European Union (EU)

The EU Machinery Safety Directive applies to Cat machines and most work tools. It requires that the "CE mark" be applied to the product and that a manufacturer's declaration be provided. The "CE mark" indicates that safety issues have been addressed by applying the appropriate safety standards in the design and manufacture of the machine. The objective of the Safety Directive is to protect operators, spectators and maintenance personnel. Caterpillar fully supports this objective.

VII. Sound Suppression

Different marketing areas have different noise emission requirements. Noise regulations usually specify limits for operators and spectators.

(a) United States

OSHA and MSHA noise-control regulations set permissible noise-exposure limits for machine operators and employees. Operator protection from machine noise can be achieved by use of factory-built cabs as offered in the Caterpillar Price List. These cabs, when properly maintained and operated with the doors and windows closed, reduce the operator sound level for an eight-hour operating period to meet the OSHA and MSHA noise-exposure limits in effect at the date of manufacture. Variables that may be encountered on the job site, such as other nearby noise sources or noise-reflecting surfaces, may reduce the allowable work hours. If this occurs, hearing protection may be required. This is especially true if a machine is not equipped with a closed cab. For example if the machine has no cab, or is being operated with the doors or windows opened.

(b) European Union

Operator sound-exposure requirements for machines in Europe are very similar to the OSHA and MSHA regulations mentioned above. In addition to operator sound-exposure requirements, most types of Cat machines are subject to European Commission regulations for exterior sound levels. Caterpillar ensures its products sold in the EU comply with the applicable noise regulations.

VIII. Replacement Parts for your Cat Machine

⚠ WARNING

When replacement parts are required for this product, Caterpillar recommends using Cat replacement parts or parts with equivalent specifications including, but not limited to, physical dimensions, type, strength and material. Failure to heed this warning can lead to premature failures, product damage, personal injury or death of persons on, or around, the product.

Nomenclature

THE CAT PRODUCT LINE

Note: Not all Sales Models are available in all countries. Engines may vary with local emission requirements. This Product Line includes production models at the moment of publishing this book.

ARTICULATED TRUCKS

725
730
730 EJ
735
740 EJ
740 GC
745

BACKHOE LOADERS

415 IL
415
416
420
420 XE
424
426 F2
428
430
432
434
440
444
450

FOREST PRODUCTS

Forest Machines

538
548/548LL
558/558LL
568

HYDRAULIC EXCAVATORS

Track Models

Mini Excavators

300.9D
300.9D VPS
301.5
301.6
301.7CR
301.8
302CR
302.7CR
303CR
303.5CR
304
305CR
305.5
306CR
306CR XTC
306.5
307.5
308CR
308CR VAB
309CR
309CR VAB
310

Small Excavators

313
313 GC
313D2
313D2 LGP
313D2 GC
315
315 GC
316 GC
317
317 GC
318D2 L

Medium Excavators

320D3 GC
320D3
320 GX
320 GC
320
323D3
323 GX
323 GC
323
325
326 GC
326
330 GC
330
333
335

Large Excavators

336 GC
336
340
345 GC
349
350
352
355
374
395

Demolition

340 SB
340 UHD
352 UHD

Wheel Models

M314
M315
M315F
M316
M317D2
M317
M318
M319
M320D2
M320
M322
M323F

HYDROMECHANICAL ATTACHMENTS

Hydraulic Hammers

H110 GC
H110 GC S
H110 S
H115 GC
H115 GC S
H115 S
H120 GC
H120 GC S
H120 S
H130 GC
H130 GC S
H130 S
H140 GC
H140 GC S
H140 S
H160 S
H180 S
H190 S
H215 S

Mobile Scrap and Demolition Shears

S3015
S3025
S3035
S2050 Straight
S3050
S2070 Straight
S3070
S2090 Straight
S3090

Multi-Processors

MP318
MP324
MP332
MP345
MP365

Contractor's Grapples

G113
G117
G120
G126
G136
G149
G174

Concrete Crushers

P315
P325
P335
P360

Secondary Pulverizers

P215
P225
P235

Demolition and Sorting Grapples

G212 GC
G213 GC
G217 GC
G312 GC
G313 GC
G314
G317 GC
G318
G318 WH
G324
G324 WH
G332
G345

Orange Peel Grapples

GSH420
GSH425
GSH440
GSH455
GSH520
GSH525
GSH555
GSM-50
GSM-60
GSV520 GC
GSV520
GSV525

MATERIAL HANDLERS

Wheeled Material Handlers

MH3022
MH3024
MH3026
MH3040

Tracked Material Handlers

MH3250
MH3260

MINING & OFF-HIGHWAY TRUCKS

770G
772G
773E
773G
775G
777E
777G
785D
785
789D
789
793D
793F
794 AC
796 AC
797F
798 AC

MOTOR GRADERS

12K
120K
120 GC
120
140K
140 GC
140
160K
150
160
14
16
18
24

PAVING PRODUCTS

Cold Planers

PM310
PM312
PM313
PM620
PM622
PM820
PM822
PM825

Rotary Mixers

RM400
RM500B

Asphalt Pavers

AP300F/AP355F
AP500F/AP555F
AP655F L
AP600F/AP655F
AP1000F/AP1055F

Pneumatic Compactors

CW12
CW16
CW34

Vibratory Asphalt Compactors

CB1.7
CB1.8
CB22B
CB24B
CB24B XT
CB32B
CC24B
CB2.5
CB2.7
CB2.9
CC2.6
CB2.5
CB2.7
CB2.9
CC2.7
CB2.5 GC
CB2.7 GC
CC2.7 GC
CB34B
CB36B
CC34B
CB4.0(03A)
CB4.4(03A)
CC4.0 (03A)
CB7
CB8
CB10
CB7 (02A)
CB10 (02A)
CCS7
CCS9
CD8
CD10
CB13
CB15
CB16

Vibratory Single Drum Soil Compactors

CS10 GC
CS11 GC
CS12 GC
CP11 GC
CP12 GC
CS34
CP34
CS44B
CP44B
CS533E
CP533E
CS54B
CP54B
CS56B
CP56B
CS64B
CS66B
CS68B
CP68B
CS74B
CP74B
CS76B
CP76B
CS79B

PIPELAYERS

PL61
PL72
PL83
PL87

SKID STEER LOADERS/COMPACT TRACK LOADERS

Skid Steer Loaders

216B3
226B3
226D3
232D3
236D3
242D3
246D3
262D3
272D3
272D3 XE

Compact Track Loaders

239D3
249D3
259D3
279D3
289D3
299D3
299D3 XE

TELESCOPIC HANDLER

TL Series

TL642
TL943
TL1055
TL1255

TH Series

TH255C
TH357
TH408
TH3510

TRACK LOADERS

953
953K
963
963K
973K

TRACK-TYPE TRACTORS

D1
D2
D3
D4
D5
D5R2
D6/D6 XE
D6R2
D6 GC
D6T
D7
D7R
D8R
D8T
D9
D9 GC
D9R
D9T
D10T2
D11
D11T

WHEEL DOZERS

814
824K
834K
844K
854K

LANDFILL COMPACTORS

816
826K
836K

SOIL COMPACTORS

815
825K

WHEEL LOADERS & INTEGRATED TOOLCARRIERS

901C2
902C2
903C2
903D
906K
906M
907K
907M
908K
908M
910
914
914K
920
920K
921E
924K
926M
930K
930M
938K
938M
950 GC
950L
950M
950M Z
962L
962M
962M Z
966 GC
966L
966M
966M XE
972L
972M
972M XE
980L
980M
980
980 XE
982
982 XE
982M
986K
988K
988K XE
990K
992
992K
993K
994K

WHEEL TRACTOR-SCRAPERS

Open Bowl Scrapers

621K
627K
631K
637K
651
657

Elevating Scrapers

623K

MINING

Draglines

8000	8200	8750

Rotary Drills

MD6200	MD6310
MD6250	MD6640

Electric Rope Shovels

7295	7495	7495 HD
7395	7495 HF	

Hydraulic Mining Shovels

6015	6020B	6030 AC	6060
6015B	6030	6040	6060 AC

Underground – Load Haul Dump Loaders

R1300G	R1700	R2900	R3000H
R1600H	R1700G	R2900G	

Underground – Mining Trucks

AD22	AD45	AD60
AD30	AD45B	AD63

NOMENCLATURE VALUE IDENTIFIER (XE, GC):

Most Cat Products are named without any value identifiers. Those models are considered Performance products that offer high production and high efficiency. However, to differentiate against other models the following value identifiers are used:

- The "XE" as an identifier for our Premium products with advanced technology where significant additional customer value is offered. The "XE" represents to our customers a family of Cat premium products that are technology leading and include a specific breakthrough technological advancement that is clearly differentiating in the industry and provides dramatic benefits to our customers' owning and operating costs.
- The "GC" is an identifier for products that are intentionally and purposefully designed for customers who value the Cat brand offering but are also the most focused on cost per hour and have the lowest cost of downtime/affordability of the Lifecycle Value segment. Customers see these machines as providing an exceptional value proposition for the applications and markets where they are designed to compete.

ARTICULATED TRUCKS

CONTENTS

Features 1-1
Ground Pressure 1-2

Features:

- **Cat® engines with ACERT™ Technology** meet U.S. EPA Tier 4 Final/EU Stage V/Japan 2014 (Tier 4 Final), or Tier 2/Stage II/Japan 2001 (Tier 2) equivalent emission standards. The four core elements of meeting these standards are:

  I) Common Rail;

  II) Electronics, ADEM™ A4;

  III) Fuel delivery, Mechanical-activated Electronic Unit Injection (MEUI™ A-C);

  IV) Air Management, Wastegate Turbocharging, Air to Air Aftercooling (ATAAC) with the proven technology of a crossflow cylinder head.

- **Cat electronically controlled transmissions** … Transmissions purpose built and designed for articulated trucks and their applications. Advanced Productivity Electronic Control Strategy (APECS) delivering smooth shifting transmissions with improved acceleration and high productivity. Providing complete integration with the engines for efficient power delivery as well as offering advanced diagnostic and troubleshooting capabilities.

- **Articulating and fully oscillating hitch** … Links front and rear frames for exceptional maneuverability and traction on uneven terrain while eliminating damaging twisting of the frames. Bolted hitch design allows optimum material choices for the cast hitch head and the hard-wearing tube. Bolted design allows easier rebuild and repair.

- **Three-point front suspension** … Three-point front suspension with long-stroke, low-pressure suspension cylinders provide unparalleled ride quality for operator comfort and higher average haul speeds. Front and rear suspension together with the hitch provide for excellent traction in all conditions.

- **Wide, long and low dump body design** … For excellent loadability and high fill factors, excellent machine stability and load retention as well as a good match for other Cat loading systems. Diverging flow design also gives excellent material ejection.

- **Standard ROPS/FOPS, low sound level cab** … Two man cab common across the range. Large cab with excellent visibility, ergonomic control layout and plentiful storage.

- **High capacity low pressure tires in single formation** … For superior traction and flotation in poor underfoot conditions.

- **Bare Chassis offerings** … For certain applications the Caterpillar OEM Solutions Group offers non-dumper/Bare Chassis arrangements.

  Bare Chassis arrangements applications could include: water, service (fuel and lube), high capacity body (waste, coal, etc.), open body (log, pipe, etc.), container carrier, hook lift, tow, cable reel, etc. Please refer to specific OEM for additional information.

Use of Ground Pressure Charts

Articulated trucks are normally equipped with wide base radial tires, for improved flotation in poor underfoot conditions. Ground pressure is a function of tire deflection and is also affected by tire penetration. The charts in this section provide a means to estimate ground pressure for 0 and 76 mm (3") tire penetration, when gross vehicle weight, axle load distribution and tire inflation pressure are known. The ground pressure charts on the following pages are based on Michelin XADN tire characteristics. Results may differ for other tread patterns.

Tire load can be calculated by the following formula:

$$\text{Tire Load} = \frac{\text{Heaviest Axle Load}}{2}$$

Example

Find the ground pressure generated by a 725C fully loaded with zero and 76 mm (3") tire penetration. The machine is equipped with standard Michelin 23.5R25 tires, inflated to the recommended pressure.

$$\text{725C Tire Load} = \frac{46\,820 \text{ kg} \times 0.34}{2} = 7959 \text{ kg}$$

$$\text{725C Tire Load} = \frac{103{,}220 \text{ lb} \times 0.34}{2} = 17{,}547 \text{ lb}$$

Note: The Front Axle of a fully loaded 725C (46,820 kg/103,220 lb machine weight) supports 34% of the load. This represents the Heaviest Axle Load.

From the tire manufacturer, inflation pressure for the 725C is 325 kPa = 3.25 bar (47 psi).

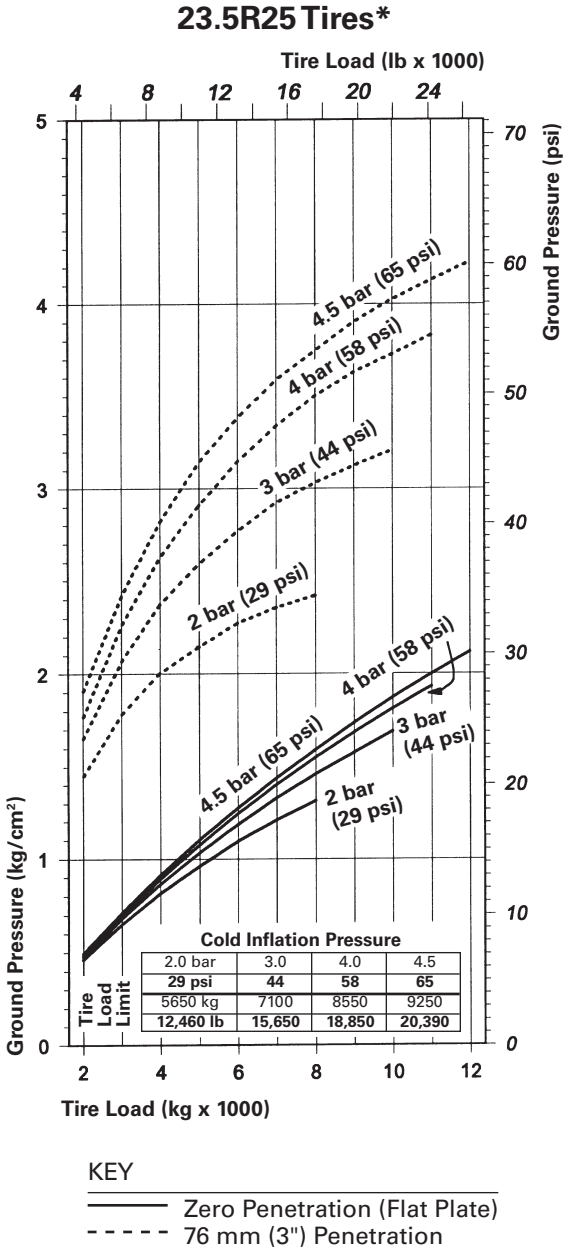
From the ground pressure chart for 23.5R25 tires, Ground pressure = 3.1 kg/cm^2 (44 psi) with zero tire penetration.

Ground pressure = 1.4 kg/cm^2 (21 psi) with 76 mm (3") tire penetration.

See the Wheel Tractor Scraper section for explanation on using:

- **Rimpull-Speed-Gradeability Curves**
- **Retarder Curves**

See Mining & Off-Highway Trucks section for Fixed Times for Hauling Units.



*Charts based on Michelin XADN tire characteristics. Results may differ for other tread patterns and/or brands. Charts are to be used to calculate ground pressure. To determine the inflation as a function of load and conditions or when loads exceed tire load limit, contact your tire manufacturer representative.

BACKHOE LOADERS

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Features:

- **Pilot operated backhoe controls** provide smooth, efficient operation and operator comfort. Thumb roller controllers provide ergonomic function of the extendible stick and/or auxiliary hydraulic circuits. Convenient pattern changer switch is within the cab. Optional on 415, 416, 428 and 434. Standard on 420 and 430. Includes E/H finger-tip actuated stabilizer controls.
- **E/H seat mounted joystick controls** are standard on 420XE, 432, 440, 444 and 450. Provides improved ergonomics, comfort, floor space and operating efficiency due to new "Dual Mode" feature. Includes E/H finger-tip actuated stabilizer controls.
- **The Cat C3.6 Direct Injection Turbocharged Aftercooled engines meet U.S. EPA & CARB Tier 4 Final and EU Stage V.** The new C3.6 is the perfect balance of compact size, reliability, and performance.
- **XT™-3 ES** hoses combined with Cat couplings and O-ring face seal fittings provide a dry, reliable machine. The F and F2 Series backhoe circuits incorporate the XT-3 ES ToughGuard™ hoses.
- **Operator station features:** Air-suspension seat is standard on all models. Adjustable tilt steering is standard except on 415 canopy configurations. Rear, door, and side windows can be fully opened for enhanced ventilation and cab roof is extended to help keep operator dry. Four-post Rollover Protective Structure (ROPS) for increased protection. Analog display with soft-key navigation/selection is standard with optional touch-screen display on certain models.
- **High performance backhoe linkage** offers 205° of bucket rotation with one pin position. Backhoe geometry creates more stick force than previous series, brought closer to the operator to pull spoil through trench. The 450F has a backhoe bucket rotation of 198°.
- **Diagonal Retention System (DRS)** standard on all factory installed buckets with weld-on tooth adapters, excluding the 450F. Bucket teeth are attached with diagonal pins rather than horizontal pins for easy exchange of bucket teeth. 450F buckets remain fitted with J225 size, horizontal fastener pins. All other buckets have bolt-on Uni-teeth.
- **Integrated lift eye** on backhoe linkage. Object handling installation available on 428, 432, 434 and 444 for EU countries.
- **Cat Cushion Swing** system smooths the swing function, improving the return-to-trench controllability.
- **The extendible stick** on 415 thru 450 models offers object clamping while extending. Serrated edges secure clamped objects. Wear pad configuration eases adjustment on all four sliding surfaces. All center pivot sticks have thumb-ready provision.
- **The hydraulic system** uses load sensing, flow sharing valves with anti-drift characteristics. Smooth, multi-function operation with the ability to have maximum lifting and digging forces at any RPM.
- **Ride Control** available as an option on all models but is standard on the 450. The ride control system smooths the ride under all job-site conditions.
- **4F/4R fully synchronized gear box** provides on-the-go shifting in all gears and on-the-go engagement of optional all-wheel drive. Maximum travel speed is 40 km/h (25 mph). Available on 415, 416, 428 and 434.
- **Powershift, with optional autoshift** is available as an option on 420, 420XE, 428, 432 and 434. Autoshift is standard on 440 and 450.
- **Lock-up torque converter** available as an option on 420, 420XE, 428, 430, 432, 434 and 440 models for improved performance and efficiency while roading.
- **Brakes** are oil immersed, multi-disc, self-adjusting, and wear surface is made of Kevlar® for long service life. Brakes are boosted on all models.
- **4WD is standard on all models.** It improves mobility and loader performance in poor traction conditions and can be engaged at any time in any operating condition. Includes 4-wheel braking effect feature.
- **Sloping, flip-open multi-access hood** allows excellent visibility to the loader working area and tilts forward for single location access to all daily service points.
- **Dry-type, radial seal air cleaner** with automatic, integrated dust ejector system provides efficient pre-separation. The two-stage air filter incorporates both air cleaner and pre-cleaner functions into a single unit mounted under the hood.

At A Glance:

Operator Comfort

- Wide Opening Windows
- Standard Air Suspension Seat
- Adjustable seat, steering column, and backhoe controls
- Easy to read and navigate displays

Versatility

- IT Loader Coupler (Optional) – All Models
- Mechanical and Hydraulic Backhoe Quick Coupler (Optional) – All Models

Control

- Hand Control DIF lock
- Brake Mode Selector Switch
- Lock up Torque Converter (Optional) – 420/420 XE/430/440
- Ride Control (Optional) – All Models

Serviceability

- Grouped Daily Service Checks
- Easy Access Cooling Package (No Tools Required)
- Battery Disconnect (Standard)
- Greaseless E-Stick with fast and easy adjustments.

Hydraulics

- Variable Displacement Piston Pump
- Flow Sharing Valves
- Mech/Pilot/EH controls
- EH Technology – 420 XE/440/450
- Full implement forces at any RPM

Safety/Security

- Transmission Neutralizer Switch
- Hydraulic Lock Out
- Back up alarm standard
- Security System (Optional) Operator Display Enabled

Road Construction & Utilities "Oil & Gas"

Examples of Basic Applications

- *Re-Surfacing Work*
  - Installing Utilities – Trenching
- *Maintenance*
  - Broom for debris clean up
  - Snow blowers to clear roadways
- *Installing Utilities – Trenching*

Governmental

Examples of Basic Applications

- *Land Management*
- *Log and Brush Clearing*
  - Grapples and Multi-purpose buckets
- *Post Holes and Fence work*
  - Use augers to bore holes
  - Forks to carry post

Agriculture

Examples of Basic Applications

- *Farm Maintenance*
  - Augers
  - MP Buckets
  - Forks for fencing
  - Tiling

General Construction

Examples of Basic Applications

- *Ground Prep*
  - Compactor level jobsite
- *Lay Drainage Pipe*
- *Installing Utilities*

DRILLS

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ROTARY BLASTHOLE DRILLS

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Key Specifications 3-4

Throughout this document, references to Tier 4 Interim/Stage IIIB/Japan 2011 (Tier 4 Interim) include U.S. EPA Tier 4 Interim, EU Stage IIIB, and Japan 2011 (Tier 4 Interim) equivalent emission standards. References to Tier 4 Final/Stage IV/Japan 2014 (Tier 4 Final) include U.S. EPA Tier 4 Final, EU Stage V, and Japan 2014 (Tier 4 Final) emission standards.

Throughout this document, references to Tier 1/Stage I include U.S. EPA Tier 1 and EU Stage I equivalent emission standards. References to Tier 2/Stage II/Japan 2001 (Tier 2) equivalent include U.S. EPA Tier 2, EU Stage II, and Japan 2001 (Tier 2) equivalent emission standards. References to Tier 3/Stage IIIA/Japan 2006 (Tier 3) equivalent include U.S. EPA Tier 3, EU Stage IIIA, and Japan 2006 (Tier 3) equivalent emission standards.

PRODUCT LINE DESCRIPTION

The Cat® Rotary Drill Family consists of the: MD6200, MD6250, MD6310, MD6380 and MD6640. With a wide range of pulldown and related weight on bit, the Cat drill family offers a maximum bit load of up to 63 975 kg (141,000 lb), to suit a wide variety of applications. They're proven to deliver efficiency, high productivity and low cost, as well as provide a comfortable environment for the operator.

With a synergy of robust structures, long-lasting systems and innovative technology, the Cat line of rotary drills has demonstrated its effectiveness and longevity in a variety of mining and quarry environments, including both soft- and hard-rock applications, as well as extreme temperature and high-altitude locations.

Equipped with low pressure compressors and hydraulic drive systems to power rotary tricone bits of various sizes and configurations, the MD6200, MD6250 and MD6310 diesel powered drills can also be configured with high pressure compressors for DTH (down-the-hole) hammer drilling. Caterpillar has the drill that delivers the optimal combination of bit load, rotary torque and onboard air to ensure maximum productivity in a wide range of applications.

PRIMARY APPLICATIONS FOR ROTARY BLASTHOLE DRILLS

MODEL	Bench Drilling	Cast Blast Drilling	Presplit Drilling	Over Burden Drilling
MD6200*	x		x	x
MD6250	x	x		x
MD6310	x	x		x
MD6380**	x	x		x
MD6640***	x	x		x

*Primary production blasthole drill, with the ability to do limited presplit drilling.
**Ultra-Class production blasthole drill, available with only low pressure compressor package.
***Ultra-Class production blasthole drill (Electric), available with only low pressure compressor package.

CAT ROTARY DRILL FEATURES

Solid Structures

The Box section frame rail construction with optimized, profiled transitions in all high stress areas provides exceptional structural durability. The A-frame and axle are integrally connected to ensure smooth load transitions. Caterpillar undercarriages have three-point oscillating suspension, GLT (grease lubricated track pins), and PPR2 (positive pin retention).

Sturdy Mast

Some models can be configured with various mast lengths to suit specific application requirements for single pass or multi pass drilling.

Designed with double-cut lacing to assure long life, mast groups are designed to ensure a torsionally strong structure reducing deflection and material fatigue.

Machined frame/mast connections eliminate custom fitment and allow for easy mast replacement or swapping of masts between drills without cutting or welding.

Jack System

Jack casing assemblies are integrally welded to the main frame providing structural strength and durability and reduce torsional flexing of the main frame.

The leveling jacks are positioned to provide excellent stability and weight over the hole.

HOBO Wrench

The variable grip Hydraulically Operated Break Out wrench is part of a highly efficient breakout system that increases speed of pipe changing operations, consequently increasing productivity.

The hydraulic break out wrench is operated from the cab thus protecting the operator.

It also reduces impact loading to the mast and rotary head during pipe changes, further increasing drill mechanical availability.

Power Group

Fast cycle times are achieved through best in class bit load, rotary head horsepower, and bailing air for maximum efficiency.

Ergonomic Cab

Cab rubber shock-mounts absorb mechanical vibration and restrict exterior noise. The operator station design integrates an ergonomic seat, multi-function joystick controls with full instrumentation, dual 254 mm (10 in) color displays, high definition touchscreen monitors, and a 12 volt power port. There is an additional display screen for cc cameras which is color, high-definition and 254 mm (10 in).

MODEL	MD6200	MD6250
Hole Diameter	• 127-200 mm **(5.0-7.87 in)**	• 152-250 mm **(6-9.8 in)**
Hole Depth	• Single-pass — 11 m **(36 ft)** • Multi-pass — Down to 47.5 m **(156 ft)**	**For short mast:** • Single-pass — 11.2 m **(36.7 ft)**, Multi-pass — Down to 53.6 m **(176.7 ft)** **For long mast:** • Single-pass — 13.6 m **(44.6 ft)**, Multi-pass — Down to 37.9 m **(124.6 ft)**
Air compressor	• Rotary — 32.6 m^3/min **(1150 ft^3/min)** @ 8.6 bar **(125 psi)** • DTH — 29.7 m^3/min **(1050 ft^3/min)** @ 24.1 bar **(350 psi)**	• Rotary — 56.6 m^3/min **(2000 ft^3/min)** @ 8.6 bar **(125 psi)** • DTH — 38.2 m^3/min **(1350 ft^3/min)** @ 34.4 bar **(500 psi)** • DTH — 42.4 m^3/min (**1500 ft^3/min)** @ 24.1 bar **(350 psi)**
Engine	• C18 @ 1800 RPM • Emissions — U.S. EPA Tier 4 Final • Power Rating (ISO 14396) — 439 kW **(589 hp)** • C18 @ 1800 RPM • Emissions — U.S. EPA Tier 2 equivalent • Power Rating (ISO 14396) — 438 kW **(587 hp)**	• C27 @ 1800 RPM • Emissions — U.S. EPA Tier 4 Final • Power Rating (ISO 14396) — 650 kW **(872 hp)** • C27 @ 1800 RPM • Emissions — U.S. EPA Tier 2 equivalent • Power Rating (ISO 14396) — 616 kW **(826 hp)**
Cooler ambient rating	• Up to 52° C **(125° F)**	• Up to 52° C **(125° F)**
Cab	• FOPS cab has 1.9 m^2 **(20.5 ft^2)** floor space	• FOPS cab has 3 m^2 **(32.3 ft^2)** floor space
Fuel Tank	• 833 L **(220 U.S. gal)**	• Primary tank is 1416 L **(374 U.S. gal)** plus expansion of +1416 L **(374 U.S. gal)** for a total of 2832 L **(748 U.S. gal)** (option)
Pipe Rack	• 9.14 m **(30 ft)** pipes	• 10.67 m **(35 ft)** pipes – for short mast • 6.1 m **(20 ft)** pipes – for long mast
Drill Pipe Capacity	• Quantity — 4 pod	• Quantity — 4 pod
Tram Speed Maximum	• high speed tram — 3.3 km/h **(2.0 mph)** low speed tram — 2.1 km/h **(1.3 mph)**	• 2.45 km/h **(1.5 mph)**
Angle Hole Drilling	• Vertical to 30° (in 5° increments) • Optional negative angle drilling package (0° to –15°, in 5° increments) for a total range of (30° to –15°)	• Vertical to 30° (in 5° increments)